

WEEK MODULE 4

1) Pseudocode
start
input N
for $i \leftarrow 1$ to N DO
 print i
End for
stop

$$T.C = O(N)$$

The loop runs exactly N times and each iteration prints one Number.

$$S.P = O(1)$$

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter the number";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        sum += i;
        cout << sum << endl;
    }
    return 0;
}
```

```
week_4.cpp X
week_4.cpp > main()
5      int N;
6      cout << "Enter a positive integer: ";
7      cin >> N;
8
9      for (int i = 1; i <= N; i++) {
10         cout << i;
11
12         if (i != N) {
13             cout << " ";
14         }
15     }
16
17     return 0;
18
PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Induleka\Desktop\collage_training> .\week_4.exe
7
1 2 3 4 5 6 7
PS C:\Users\Induleka\Desktop\collage_training>
```

2) Pseudocode

Start

Input N

sum $\leftarrow$ 0

for $i \leftarrow 1$ to N Do

 sum $\leftarrow$ sum + i

End for

print Sum

stop

T.C = $O(N)$

The loop runs exactly N times, and each iteration performs one addition.

S.C = $O(1)$

week_4.cpp X

week_4.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int N;
6      cin >> N;
7
8      long long sum = 0;
9
10     for (int i = 1; i <= N; i++) {
11         sum = sum + i;
12     }
13
14     cout << sum;
15
16     return 0;
17 }
```

PROBLEMS 7

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
● PS C:\Users\Induleka\Desktop\collage_training> g++ week_4.cpp -o week_4
● PS C:\Users\Induleka\Desktop\collage_training> .\week_4.exe
5
15
PS C:\Users\Induleka\Desktop\collage_training> 
```

3) Pseudocode

```
START
Input n, B
result ← 1
for i ← 1 to B Do
    result ← result × A
End for
Print result
STOP
```

T.C = $O(B)$

loops runs exactly B times and each iteration performs one multiplication

S.C = $O(1)$

```
week_4.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int A, B;
6      cin >> A >> B;
7
8      long long result = 1;
9
10     for (int i = 1; i <= B; i++) {
11         result = result * A;
12     }
13
14     cout << result;
15
16     return 0;
17 }
```

PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\Induleka\Desktop\collage_training> g++ week_4.cpp -o week_4
● PS C:\Users\Induleka\Desktop\collage_training> .\week_4.exe
5
5
3125
❖ PS C:\Users\Induleka\Desktop\collage_training> 
```